

Network and System Management

Domain Wanderers

Level 3: Step 2 - Assignment Report

Submitted by:

Student: a57540 (Hasib) & a61521 (Refat)

Introduction

This assignment focuses on implementing enterprise-level network infrastructure using Windows Server 2016 Active Directory Domain Services and exploring network file sharing protocols. The primary objective is to gain hands-on experience in managing user authentication, computer management, and file sharing across networked systems—skills that are fundamental in modern IT environments.

Throughout this project, we configured a complete domain infrastructure including a domain controller, multiple workstations, and examined both Windows (SMB) and Linux (NFS) file sharing mechanisms. This report documents our implementation process, presents theoretical understanding of key concepts, and demonstrates the practical application of domain-based network management.

Question 1: Active Directory in Windows Server

Role of Active Directory

Active Directory (AD) serves as the backbone of Windows-based network management, providing a centralized directory service that transforms how organizations handle their IT infrastructure. Rather than managing each computer independently, Active Directory creates a unified system where administrators can control users, computers, and resources from a single location.

Core Functionality

At its foundation, Active Directory operates as a hierarchical database that stores information about network objects. When implemented in Windows Server environments, it enables administrators to create a structured organization of users, groups, computers, and other resources. This structure follows a logical tree-based hierarchy, making it intuitive to manage even complex networks with thousands of devices.

User Management Benefits

One of the most significant advantages of Active Directory is centralized user authentication. Instead of creating separate user accounts on each computer, administrators establish a single domain account that works across all network resources. When a user logs in at any domain-joined computer, Active Directory verifies their credentials and grants access to appropriate resources based on their permissions. This approach dramatically simplifies account management—changing a password or updating permissions happens once in Active Directory, and the changes immediately apply network-wide.

Computer Management at Scale

For organizations with numerous computers, Active Directory provides powerful management capabilities through Group Policy Objects (GPOs). These policies allow administrators to configure security settings, software installations, desktop environments, and system behaviors across hundreds or thousands of machines simultaneously. Rather than visiting each computer individually, an administrator can define a policy once and have it automatically applied to all targeted systems. This capability becomes invaluable when managing large networks, ensuring consistency and reducing the time required for system administration.

Security and Access Control

Active Directory enhances network security through its sophisticated permission system. Administrators can organize users into security groups and assign permissions to these groups rather than individual users. This group-based approach simplifies permission management and reduces errors. Additionally, AD supports features like roaming profiles, which allow users to access their personalized settings from any computer on the network, and folder redirection, which ensures important user data is stored on network servers rather than local machines.

Practical Applications

In practical terms, Active Directory transforms network management from a computer-by-computer task into a streamlined, policy-driven process. When a new employee joins an organization, their account is created once in Active Directory, granting them appropriate access to email, file shares, applications, and other resources based on their role. Similarly, when someone leaves, disabling their AD account immediately revokes access across the entire network. This centralized control proves essential for maintaining security and operational efficiency in organizations of any size.

Question 2: NFS Security Concern - SUID Permissions

Understanding the Security Risk

Network File System (NFS) presents a particular security challenge related to SUID (Set User ID) permissions when accessed by client systems. To understand why this matters, we need to examine how SUID permissions work and why they become problematic in networked environments.

What is SUID?

The SUID permission is a special file attribute in Unix-like systems that allows users to execute a program with the permissions of the file's owner rather than their own permissions. For example, the 'passwd' command needs to modify system password files that regular users cannot normally access. By setting the SUID bit on this program and making it owned by root, any user can run it, but it executes with root privileges, allowing it to update the password file.

The NFS Security Problem

The security concern arises when an NFS server shares a filesystem that clients can mount. On the client side, the root user has complete control over their local system. If a malicious root user on a client machine can write to the NFS share, they could potentially upload a program, set the SUID bit on it, and change its ownership to root. When other users then access this shared file from their own systems, they might unknowingly execute this program, which would run with root privileges—exactly what the attacker intended.

This scenario creates a privilege escalation pathway. Even though users on other client machines don't have root access on their own systems, they could inadvertently execute programs with elevated privileges through the shared NFS filesystem. This fundamentally breaks the security model of Unix-like systems, where privilege elevation should be strictly controlled.

Prevention Measures

Several protective measures can mitigate this risk:

- **Root Squashing:** This is the most common and effective defense. NFS servers can be configured with 'root_squash' option in /etc/exports, which automatically maps the client's root user (UID 0) to the nobody user on the server. This means even if root on the client tries to set SUID permissions, those permissions are applied as the nobody user, not as root, effectively neutralizing the threat.
- **No SUID Option:** The 'nosuid' mount option instructs the client system to ignore SUID and SGID bits on files from the NFS share. Even if these bits are set on the server, they won't be honored when accessed through NFS. This can be configured either on the server side (in /etc/exports) or enforced by clients during mounting.

- **Read-Only Mounts:** When appropriate, sharing filesystems as read-only eliminates the ability to upload malicious files altogether. This works well for resources that clients only need to read, such as shared software installations or reference documentation.
- **Network Segmentation:** Restricting NFS shares to trusted networks and specific IP addresses reduces the attack surface. By limiting which systems can mount NFS shares, administrators minimize the risk of malicious clients accessing the filesystem.

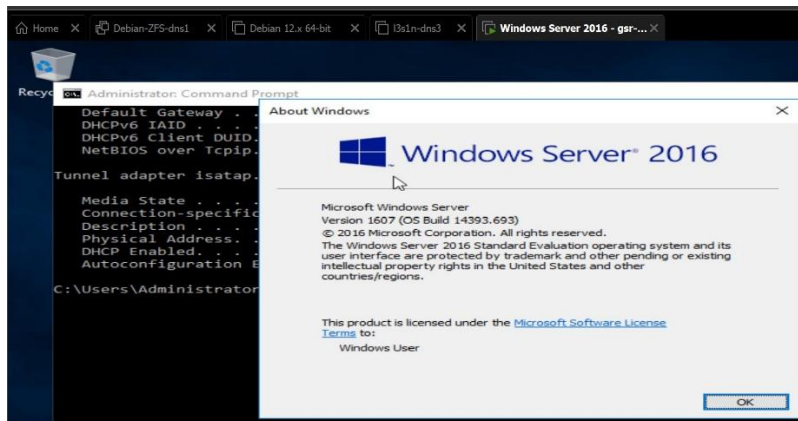
In practice, a combination of these measures provides the best security. Most NFS deployments should use root squashing by default, apply nosuid where possible, and carefully consider which systems truly need write access to shared filesystems.

Question 3: Windows Server 2016 Domain Controller Configuration

This section documents the step-by-step process of setting up a Windows Server 2016 system as an Active Directory Domain Controller for the `gsr.ipb.local` domain.

3(a) - Windows Server 2016 Installation

We began by creating a new virtual machine and installing Windows Server 2016 Standard Evaluation with Desktop Experience. This version provides the full graphical interface necessary for learning and administration tasks.



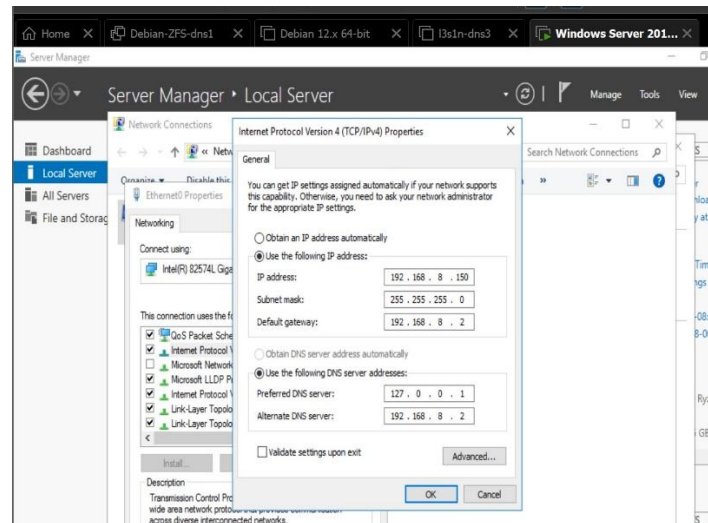
Screenshot 1: Windows Server 2016 version information showing Standard Evaluation (Desktop Experience)

3(b) - Static IP Configuration

After installation, we configured a static IP address for the server. Domain Controllers require static IP addresses to ensure consistent DNS resolution and domain services. Our network uses the 192.168.8.0/24 subnet, and we assigned the server IP address 192.168.8.150.

Network Configuration:

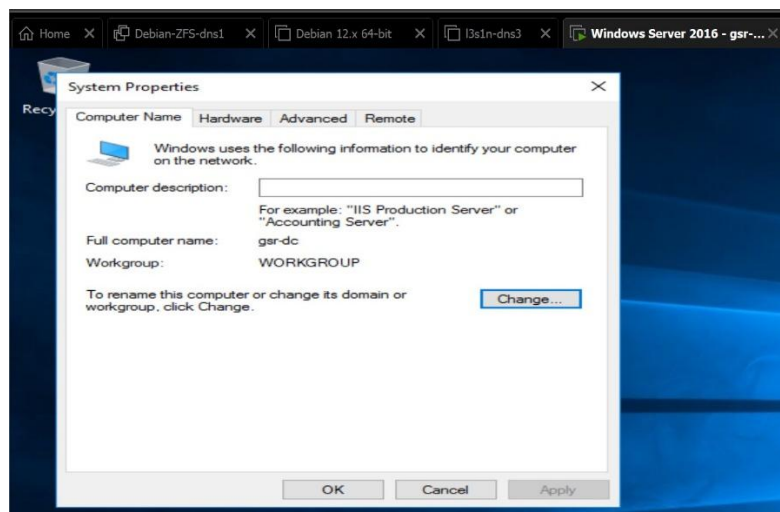
- IP Address: 192.168.8.150
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.8.2
- Preferred DNS: 127.0.0.1 (localhost)
- Alternate DNS: 192.168.8.2



Screenshot 2: ipconfig /all output showing static IP configuration

3(c) - Computer Rename

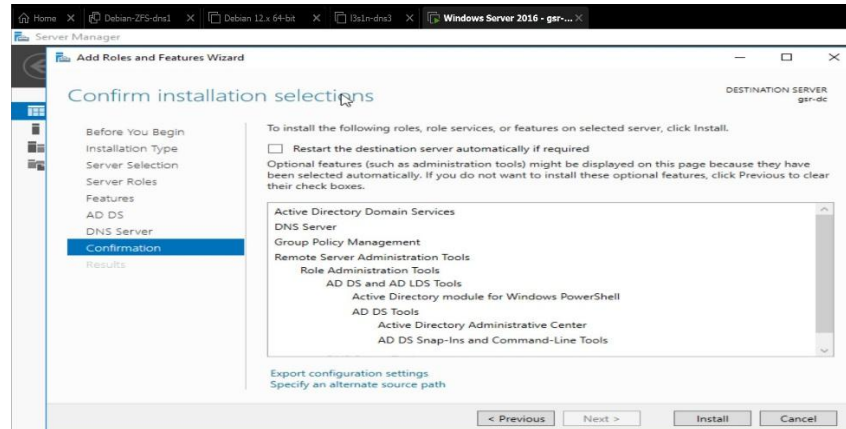
We renamed the server to 'gsr-dc' (GSR Domain Controller) to clearly identify its role in the network infrastructure. This naming convention helps administrators quickly understand the server's purpose.



Screenshot 3: System Properties showing computer name changed to gsr-dc

3(d) - Installing AD DS and DNS Server Roles

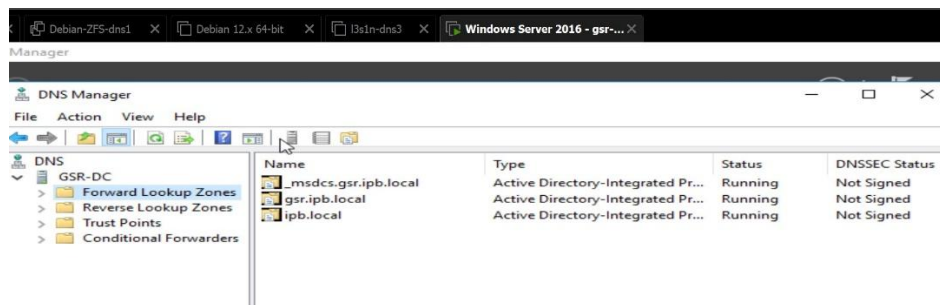
Using Server Manager, we added two critical server roles: Active Directory Domain Services and DNS Server. These roles work together—AD DS provides the domain services while DNS ensures that computers can locate domain controllers and other network resources by name.



Screenshot 4: Server Manager showing successful installation of AD DS and DNS Server features

3(e) - DNS Forward Lookup Zone Creation

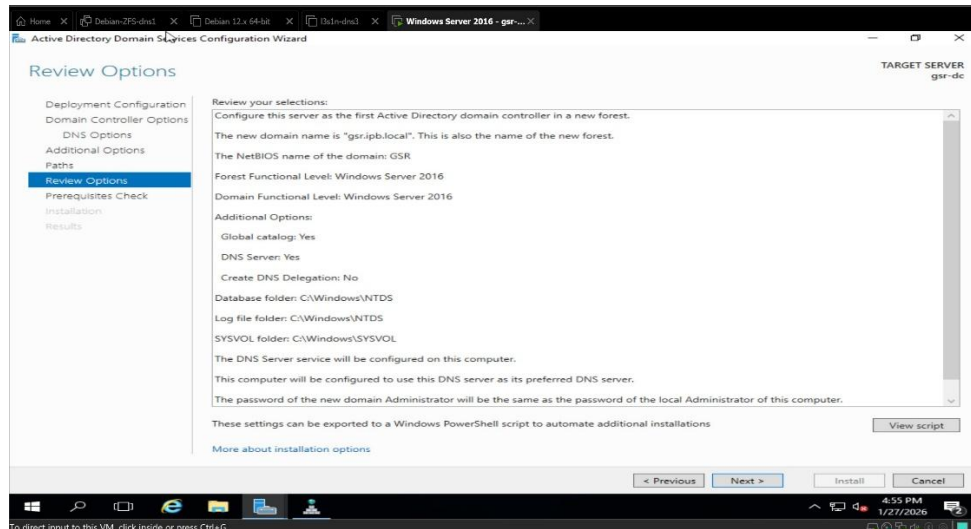
We created the ipb.local forward lookup zone in DNS Manager. This zone allows DNS to resolve domain names to IP addresses within our network infrastructure.



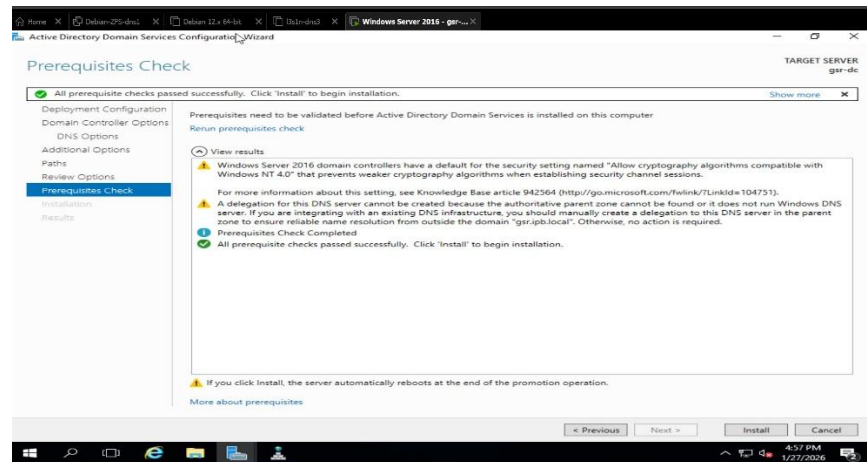
Screenshot 5: DNS Manager showing ipb.local and gsr.ipb.local forward lookup zones

3(f) - Domain Controller Promotion

The next critical step involved promoting the server to a Domain Controller for a new forest named gsr.ipb.local with NetBIOS name GSR. This process configures the server to manage authentication, authorization, and directory services for the entire domain. During promotion, we encountered an expected DNS delegation warning which we safely ignored as per the assignment instructions.



Screenshot 6: Active Directory Domain Services Configuration Wizard showing domain configuration



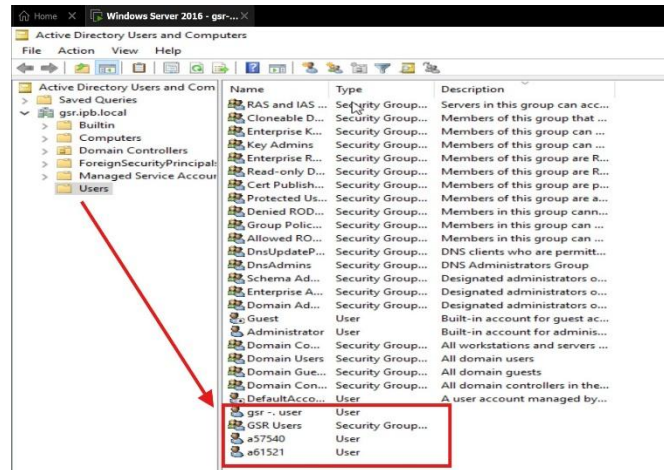
Screenshot 7: Prerequisites check passed before DC promotion

3(g) - Creating Users and Security Groups

After the Domain Controller came online, we used Active Directory Users and Computers to create our domain objects. We established a security group called 'GSR Users' and created three user accounts:

- a57540 (Hasib) - added to GSR Users group
- a61521 (Partner) - added to GSR Users group
- gsr-user - not added to GSR Users (only in Domain Users)

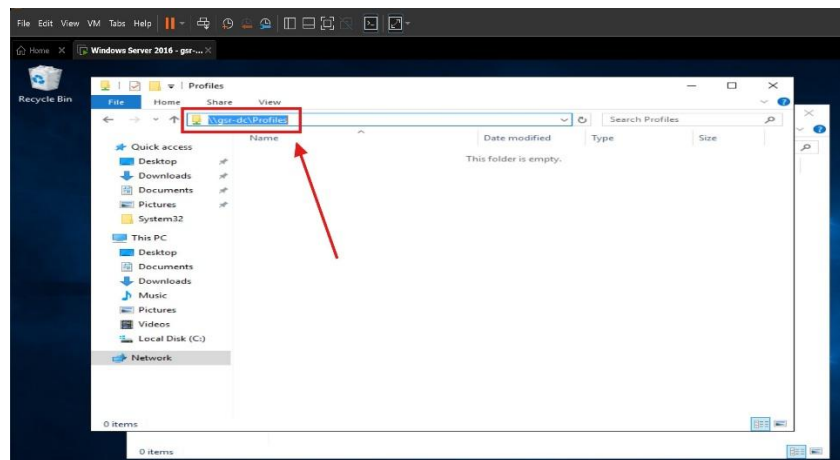
All three users are automatically members of the Domain Users group, but only the first two belong to GSR Users. This distinction becomes important later when implementing roaming profiles.



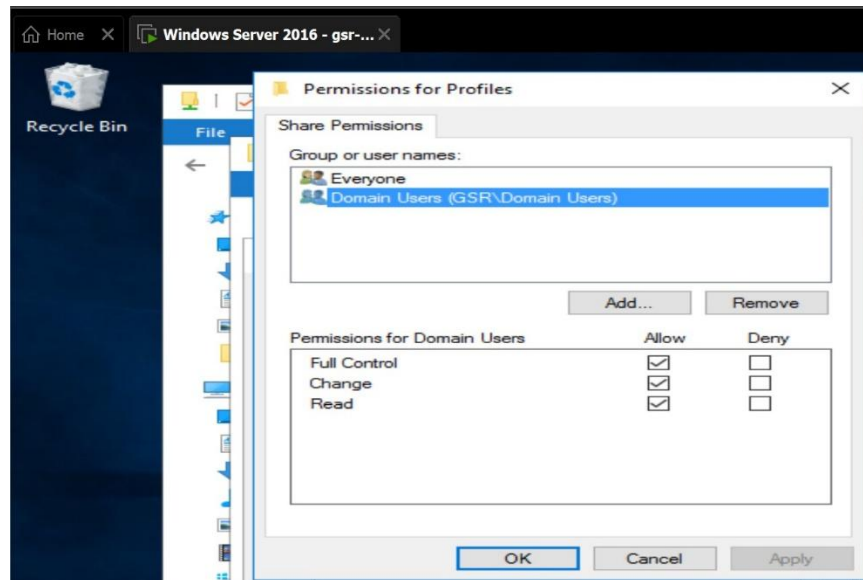
Screenshot 8: Active Directory Users and Computers showing all users, GSR Users group membership, and gsr-user with Domain Users only

3(h) - Creating and Sharing the Profiles Folder

We created a folder structure at C:\Shared\Profiles and configured it as a network share accessible at \\gsm-dc\profiles. This shared location will store roaming user profiles, allowing users to access their personalized settings from any domain-joined computer.



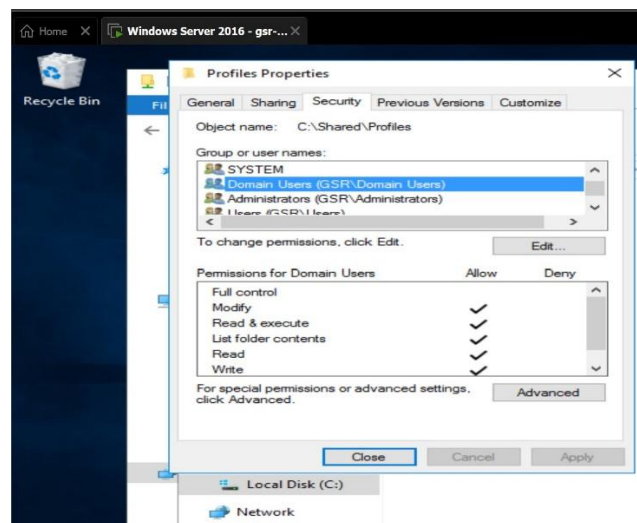
Screenshot 11: Advanced Sharing settings showing the Profiles folder shared



Screenshot 12: Share permissions showing Domain Users with Full Control

3(i) - Configuring NTFS Permissions

Beyond share permissions, we also configured NTFS security permissions on the Profiles folder. We granted Domain Users group the Modify permission, allowing users to create and modify their profile folders while maintaining appropriate security. This dual-layer permission system (share + NTFS) ensures proper access control.

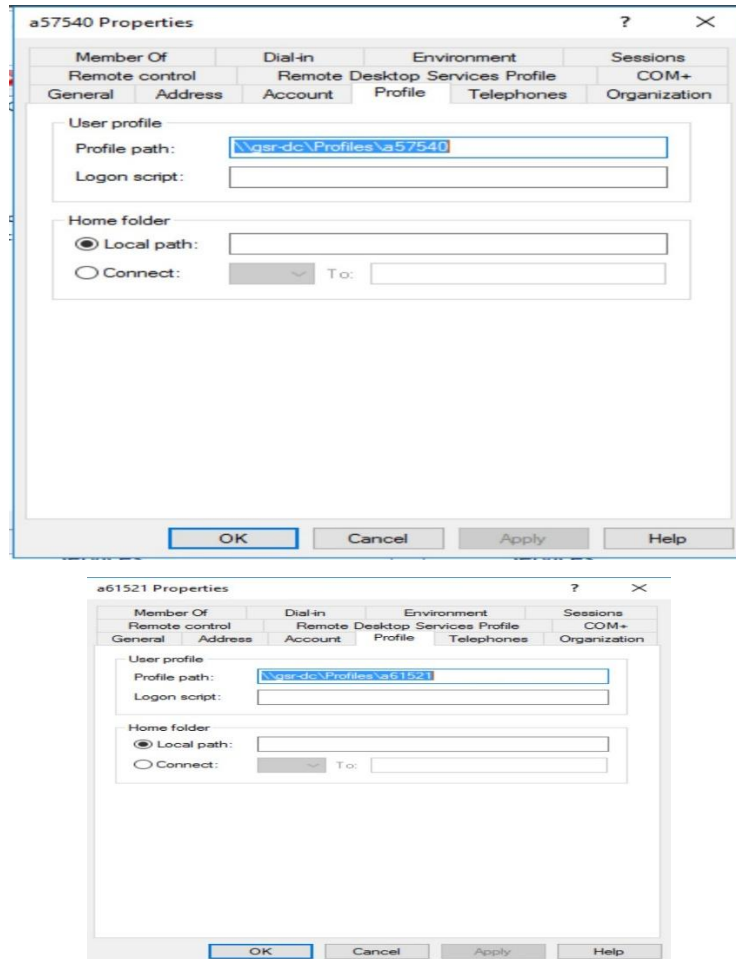


Screenshot 13: Security tab showing Domain Users with Modify permissions on Profiles folder

3(j) - Configuring Roaming Profiles

The final configuration step involved setting up roaming profiles for users in the GSR Users group. We edited each user's properties in Active Directory, specifying the profile path as \\gsr-dc\profiles\%USERNAME%. The %USERNAME% variable ensures each

user gets their own profile subfolder. Notably, we left this field empty for gsr-user, meaning that account will use a local profile instead of a roaming one.



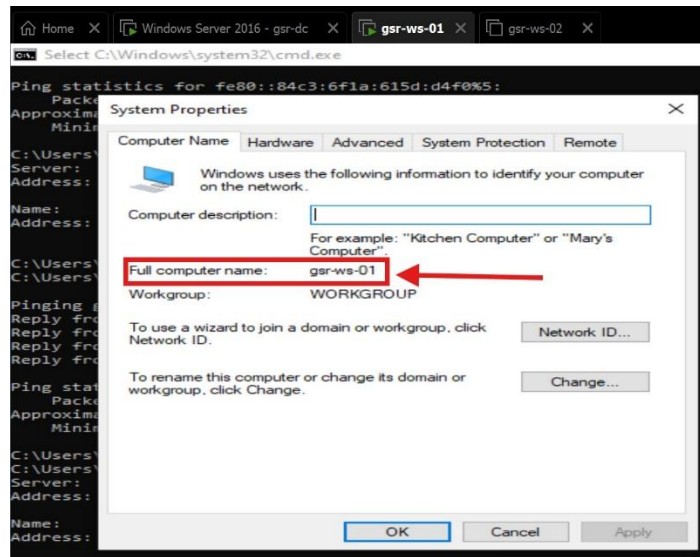
Screenshot 14: User a57540 Profile tab showing roaming profile path configured

Question 4: Windows 10 Workstation Configuration and Domain Join

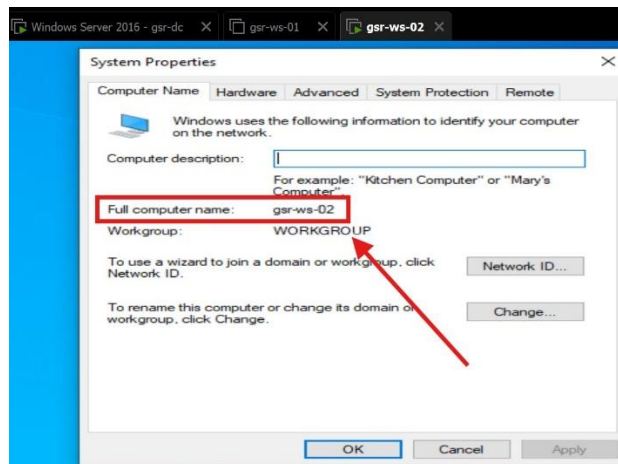
With the Domain Controller fully configured, we proceeded to set up two Windows 10 workstations and integrate them into the domain environment.

4(a) - Renaming Workstations

We renamed the two Windows 10 virtual machines to gsr-ws-01 and gsr-ws-02 respectively. This consistent naming scheme helps identify the computers within the network and follows standard IT naming conventions.



Screenshot 15: System Properties of gsr-ws-01 showing computer name



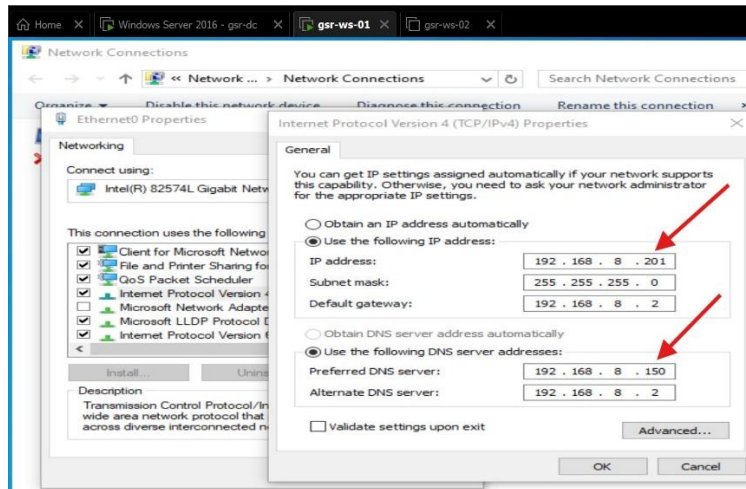
Screenshot 16: System Properties of gsr-ws-02 showing computer name

4(b) - DNS Configuration

Before joining the domain, we configured both workstations to use the Domain Controller as their primary DNS server. We assigned static IP addresses to avoid potential issues with DHCP-assigned addresses changing.

Network Configuration for gsr-ws-01:

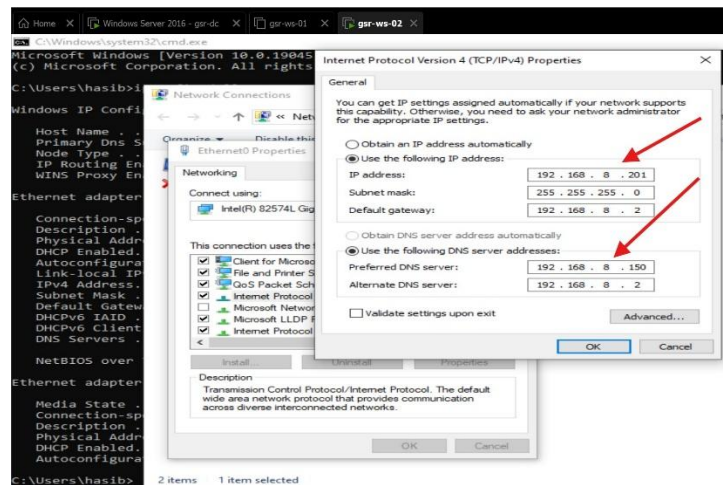
- IP Address: 192.168.8.201
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.8.2
- Preferred DNS: 192.168.8.150 (gsr-dc)
- Alternate DNS: 192.168.8.2



Screenshot 17: TCP/IPv4 Properties for gsr-ws-01 showing DNS configuration

Network Configuration for gsr-ws-02:

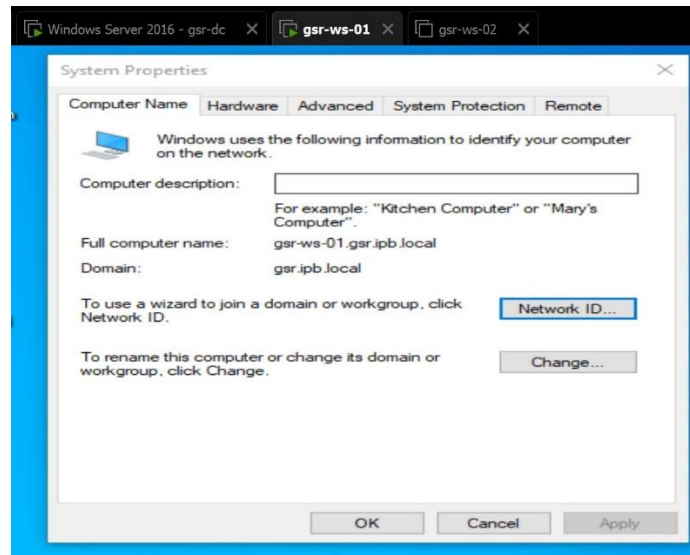
- IP Address: 192.168.8.202
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.8.2
- Preferred DNS: 192.168.8.150 (gsr-dc)
- Alternate DNS: 192.168.8.2



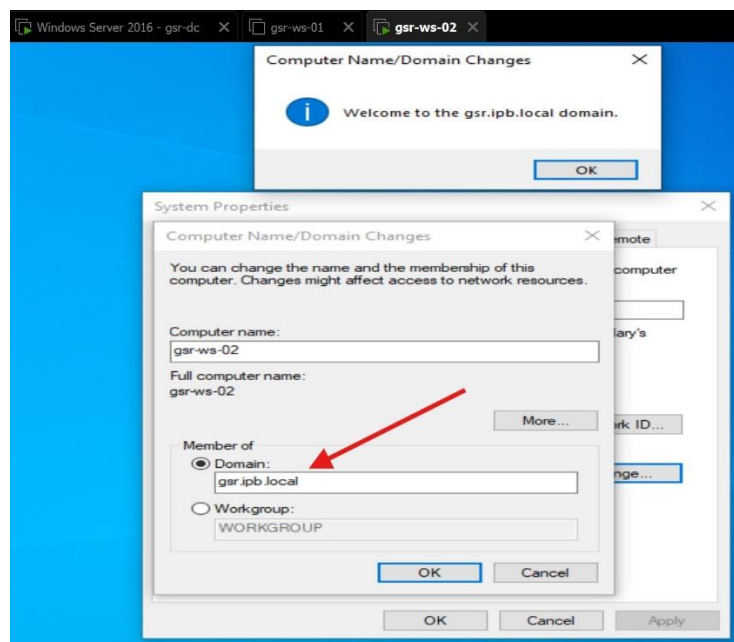
Screenshot 18: TCP/IPv4 Properties for gsr-ws-02 showing DNS configuration

4(c) - Joining the Domain

With DNS properly configured, we joined both workstations to the gsr.ipb.local domain. The process involved providing domain administrator credentials, and upon successful authentication, each computer received confirmation of joining the domain. After restarting, both systems became fully integrated domain members.



Screenshot 19: System Properties of gsr-ws-01 showing domain membership (gsr.ipb.local)



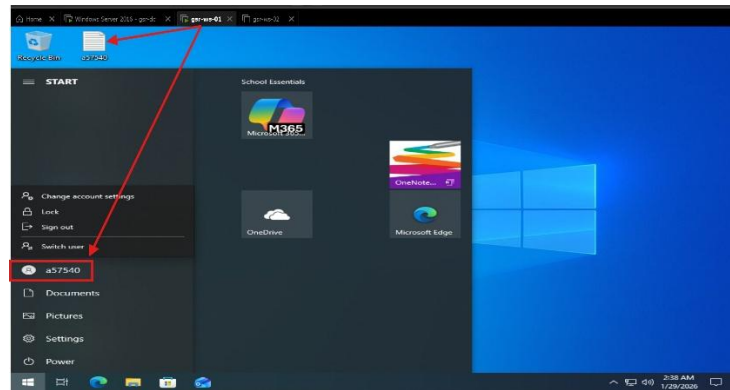
Screenshot 20: System Properties of gsr-ws-02 showing domain membership (gsr.ipb.local)

4(d) - Testing User Logins and Roaming Profiles

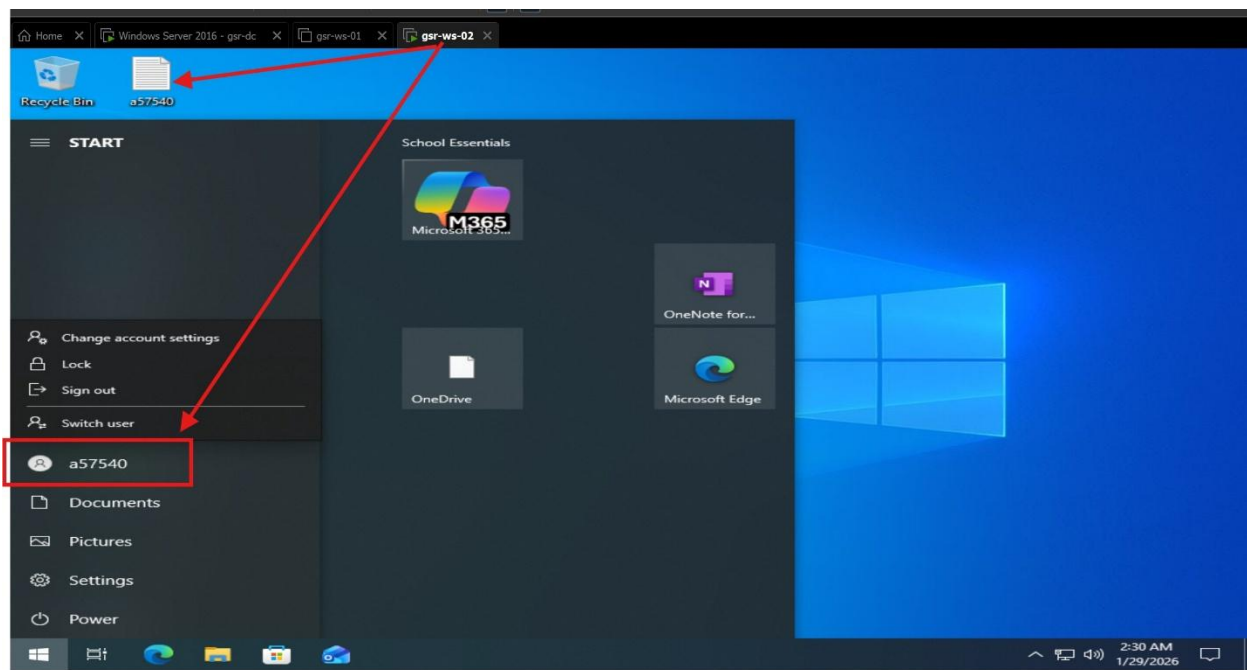
The final demonstration involved testing that all domain users could log into both workstations and verifying that roaming profiles functioned correctly for users in the GSR Users group.

Roaming Profile Test

We logged into gsr-ws-01 as user a57540 and made several personalized changes: modified the desktop wallpaper, created a test file on the desktop, and pinned an application to the taskbar. After logging out and then logging in as the same user on gsr-ws-02, all these customizations appeared exactly as configured on the first workstation. This demonstrated that the roaming profile successfully synchronized between computers, storing the user's settings on the network share and applying them regardless of which computer they use.

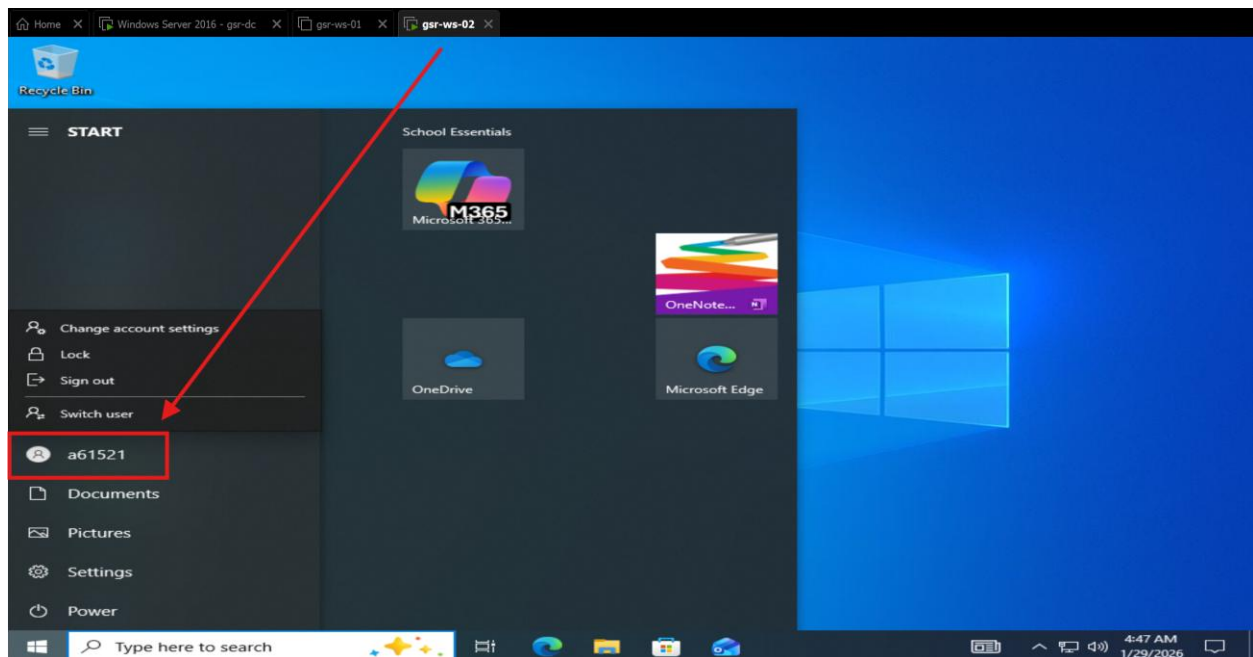


Screenshot 21: User a57540 logged into gsr-ws-01 with custom wallpaper and test file



Screenshot 22: User a57540 logged into gsr-ws-02 showing same customizations

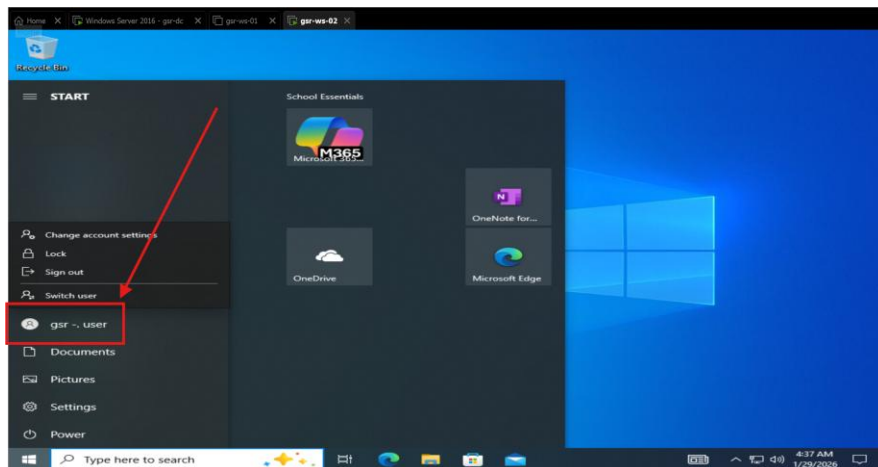
Local Profile: Additional Screenshot



Screenshot 23: User a61521 logged into gsr-ws-02 showing without customization customizations

Local Profile Test

To contrast with roaming profiles, we also logged in as gsr-user, who was not configured with a roaming profile. This account used a local profile on each machine, meaning customizations made on one workstation did not appear on the other. This confirmed that our roaming profile configuration correctly applied only to the intended users.



Screenshot 24: User gsr-user logged in showing default/local profile without roaming Settings

Question 5: NFS Export Configuration

This question explores how to configure NFS (Network File System) exports on a Debian Linux server to share directories with network clients.

5(a) - Sharing Read/Write to Specific Subnet

To share the `/home/gsr` directory with read-write access to all computers in the `192.168.122.*` subnet, we would add the following line to `/etc/exports`:

```
/home/gsr    192.168.122.0/24(rw,sync,no_subtree_check)
```

Let me break down what each component means:

- `/home/gsr` - This is the directory path we're sharing from the server
- `192.168.122.0/24` - This specifies the subnet allowed to access the share, covering all IP addresses from `192.168.122.1` to `192.168.122.254`
- `rw` - Grants read-write access, allowing clients to both read files and create or modify content
- `sync` - Ensures data is written to disk before confirming the write operation, preventing data loss if the server crashes
- `no_subtree_check` - Disables subtree checking for better reliability, particularly important when the exported directory is not a complete filesystem

5(b) - Sharing Read-Only to Everyone

To share the same directory with read-only access to any client on the network, the configuration would be:

```
/home/gsr    *(ro,sync,no_subtree_check)
```

The asterisk (*) acts as a wildcard, matching any IP address or hostname that attempts to connect. The 'ro' option restricts access to read-only, preventing clients from modifying, creating, or deleting files. This configuration is useful for sharing resources like documentation or software installations that users need to access but shouldn't modify.

5(c) - Applying Configuration Changes

After modifying the `/etc/exports` file, changes don't take effect automatically. To apply the new configuration without restarting the entire NFS service, we use the `exportfs` command:

```
exportfs -ra
```

The '-r' flag tells `exportfs` to re-export all directories, reading the updated `/etc/exports` file. The '-a' flag applies the command to all exports. This combination refreshes the NFS export table with your changes while maintaining existing client connections where possible, making it much cleaner than restarting the service.

Question 6: NFS Client Mount Configuration

This question addresses how to configure automatic mounting of NFS shares on client systems using the `/etc/fstab` file.

fstab Entry for Automatic NFS Mount

To automatically mount the `/home/gsr` share from the server at `192.168.122.180` to a local directory at boot time, we would add this line to `/etc/fstab`:

```
192.168.122.180:/home/gsr    /mnt/gsr    nfs    defaults,_netdev    0
0
```

Understanding each field in this configuration:

- `192.168.122.180:/home/gsr` - The first field identifies the remote resource, combining the server's IP address with the exported directory path
- `/mnt/gsr` - The second field specifies the local mount point where the remote filesystem will be accessible. This directory must exist before mounting
- `nfs` - The third field declares the filesystem type as NFS
- `defaults,_netdev` - The fourth field contains mount options. 'defaults' applies standard mount options (`rw`, `suid`, `dev`, `exec`, `auto`, `nouser`, `async`). The '`_netdev`' option is particularly important for NFS—it tells the system this is a network device, ensuring the mount only happens after network services are available during boot
- `0` - The fifth field (dump) should be 0 for network filesystems, as they can't be backed up using the dump utility
- `0` - The sixth field (fsck pass number) should also be 0 for network filesystems, as they shouldn't be checked by fsck at boot

After adding this entry to `/etc/fstab`, the share will automatically mount when the system boots. You can also mount it immediately without rebooting by running '`mount -a`', which mounts all filesystems listed in `fstab` that aren't already mounted.

Conclusion

Through this assignment, we successfully implemented a complete Active Directory domain infrastructure and explored network file sharing technologies. The practical experience of configuring a Domain Controller, managing user accounts and groups, setting up roaming profiles, and joining workstations to the domain provided valuable insights into enterprise network management.

The hands-on work reinforced theoretical concepts about centralized authentication and authorization. We saw firsthand how Active Directory simplifies user management—creating accounts once, configuring permissions through groups, and enabling users to access resources from any domain-joined computer. The roaming profiles feature particularly demonstrated the power of network-based solutions, allowing users to maintain consistent environments regardless of their physical location.

On the theoretical side, examining NFS security concerns and configuration deepened our understanding of network file sharing protocols. Learning about the SUID permission vulnerability and its mitigation strategies highlighted the importance of security considerations when designing network infrastructure. The practical NFS configuration exercises complemented the Windows-focused implementation work, showing different approaches to solving the same fundamental problem of sharing files across networks.

The skills developed through this assignment—domain controller setup, user management, security configuration, and network file sharing—form the foundation of modern IT infrastructure management. These competencies are directly applicable to real-world scenarios, from small business networks to large enterprise environments. The experience gained will prove valuable in future system administration tasks and provides a solid base for further exploration of advanced topics like Group Policy, distributed file systems, and complex multi-domain environments.